

Project Initialization and Planning Phase

Date	28 June 2026
Team ID	XXXXXX
Project Name	Smart Agricultural Production Optimization Engine (OptiCrop)
Maximum Marks	3 Marks

Define Problem Statements (Customer Problem Statement Template):

The Smart Agricultural Production Optimization Engine (OptiCrop) is designed to assist farmers, agricultural researchers, and policymakers in making informed decisions using Machine Learning. The system analyzes soil nutrients and environmental conditions to recommend the most suitable crop, thereby improving productivity, reducing resource wastage, and promoting sustainable farming.

Example:

I am	I'm trying to	But	Because	Which makes me feel
A farmer cultivating agricultural land	Select the most suitable crop for maximum yield and better farming efficiency	I am unsure which crop is best for my soil and climate conditions	Soil nutrients, rainfall, temperature, humidity, and pH vary from season to season	Confused about choosing the right crop and worried about productivity and income

Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	A farmer cultivating agricultural land	Select the most suitable crop for maximum yield	I am unsure which crop is best for my soil and climate conditions	Soil nutrients, rainfall, temperature, humidity, and pH vary from season to season, making crop selection difficult	Confused about choosing the right crop and concerned about productivity and income.